Level: Bachelor | Semester: Fall | Year : 2022
Programme: BE | Full Marks: 100
Course: Microprocessors | Pass Marks: 45
Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

***Attempt all the questions.***

1. a) Define Microprocessor along with its applications. Differentiate Harvard and Von-Neumann architecture on the basis of storage. 7
   b) What are the types of instruction depending upon word size? Explain different type of addressing modes of 8085 microprocessor. 8
2. a) Write an 8085 program to find smallest number among array of numbers. 8
   b) Define T-state. Draw the labelled timing diagram of the instruction LXI H, 5051H. 7
3. a) Design an Addressing Decoding circuit to interface 16KB ROM chips with 8085 microprocessors. 8
   b) Draw a well labelled architecture of 8086 microprocessor and discuss about BIU and EU. 7
4. a) Explain about IVT of 8086. 7
   b) What are serial and parallel communication? Explain basics of RS-232. 8
5. a) Explain the working of DMA. Mention its advantages over programmed and interrupt driven I/O. 8
   b) Write an ALP in 8086 to display POKHARA UNIVERSITY without using 09h function. 7
6. a) What is the Asynchronous serial data communication? Compare serial and parallel communication. Draw the Block diagram of 8251A USART and explain functionally in detail. 8
   b) What is the importance of DAA instruction? Explain it with an appropriate example along with the necessary calculations. 7
7. Write short notes on: **(Any two)**
   a) Memory mapped I/O vs I/O mapped I/O
   b) Address Bus ,Data Bus and Control Bus
   c) Flags in 8086



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